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Highly selective 4-alkynoic acids synthesis *via* iron-mediated complete inversion of stereogenic carbon centers†

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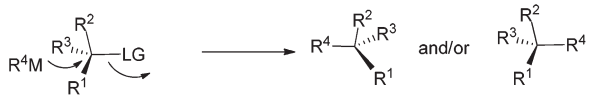
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An Fe-catalyzed highly regio- and diastereoselective reaction of 4-ethynyloxetan-2-ones with different Grignard reagents affording alkynes **3** in good yields with absolute configuration of the β-carbon atom completely inverted has been developed. The optically active acids may be easily converted to esters, which may easily undergo desilylation and highly diastereoselective allenylation to afford optically active 4,5-allenoates with two chiral centers and one axial allene chirality.

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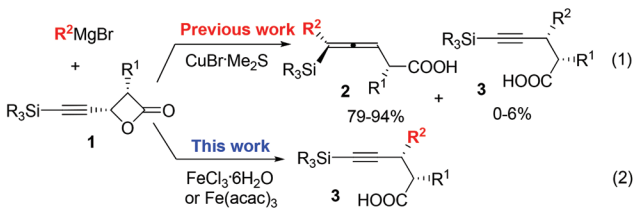
Introduction

As is well known, the stereoselectivity in the reaction of a chiral carbon center bearing a leaving group such as halide, tosylate, or tosylamide with a hard carbon nucleophile, such as Grignard reagents, lithium reagents, or lithium cuprates, is a basic topic of current interest with either high stereoselectivity for inversion or partial racemization.[1–9] Although iron compounds or complexes have been recently reported to be very efficient catalysts for many organic transformations,[10] the $FeCl_3$-catalyzed reactions of ArMgX with optically active alkyl bromides proceed with complete racemization.[11] Surprisingly, such reactions with the more readily available yet stable acetates are quite limited (Scheme 1).[12]



Scheme 1 Inversion of a chiral carbon center.

On the other hand, in 2000, Nelson *et al.* reported the Cu(I)-catalyzed S_N2' type coupling reaction of 4-(trimethylsilylethynyl) oxetan-2-one **1** with alkyl Grignard reagents for the synthesis of β-allenoic acid derivatives **2** together with the formation of γ-propargylic acids **3** as the minor products (eqn (1));[13–15] Fürstner *et al.* reported the $Fe(acac)_3$-catalyzed reaction of Grignard reagents with 1-alkynyl epoxides affording 2,3-allenols[16] and the $[Fe(MgX)_2]_n$-catalyzed coupling reaction of primary propargylic bromides with Grignard reagents affording alkynes.[17–19] Herein, we wish to report the highly regio- and stereo-selective Fe-catalyzed reaction of 4-(trimethylsilylethynyl)oxetan-2-one **1** with Grignard reagents affording 4-alkynoic acids **3** with absolute configuration of the chiral center being completely inverted.



Results and discussion

During the study of the reaction of *cis*-3-methyl-4-(trimethylsilylethynyl)oxetan-2-one **1a** with 3 equiv. of MeMgCl, it was observed that the reaction catalyzed by 10 mol% of $Fe(acac)_3$ at −78 °C afforded 2,3-dimethyl-5-(trimethylsilyl)pent-4-ynoic acid **3a** as the major product with a selectivity of **3a**/**2a** = 97/3 and dr value of **3a** being about 100/0 as judged by the ^1H NMR analysis of the crude reaction mixture (entry 1, Table 1). Other solvents such as Et_2O and toluene led to a lower yield and selectivity (entries 2 and 3, Table 1). $FeCl_3{\cdot}6H_2O$ demonstrated a similar results and the loading of catalyst could be reduced to 5 mol% (entries 6 and 7, Table 1). Anhydrous $FeCl_3$ catalyzed this transformation affording **3a** in 61% yield together

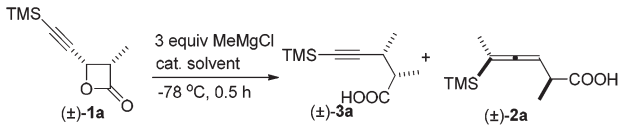
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† Electronic supplementary information (ESI) available: Experimental procedures and detailed characterization data for all new compounds. CCDC 940135. For ESI and crystallographic data in CIF or other electronic format see DOI: 10.1039/c3qo00066d

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with **2a** in 3% yield (entry 8, Table 1). Iron salt is vital to the coupling reaction: neither **3a** nor **2a** was provided from the blank test and the control experiment with CuCl as catalyst (entries 9 and 10, Table 1).

Table 1 Effects of catalyst and solvent on the coupling of MeMgCl with **1a**[a]



Entry	Cat. (mol%)	Solvent	Yield of **3a**[b] (%)	**3a**/**2a**
1	$Fe(acac)_3$ (10)	THF	86	97/3
2	$Fe(acac)_3$ (10)	Et_2O	79	95/5
3	$Fe(acac)_3$ (10)	Toluene	77	95/5
4	$Fe(acac)_2$ (10)	THF	87	97/3
5[c]	$FeCl_2$ (10)	THF	87	98/2
6	$FeCl_3{\cdot}6H_2O$ (10)	THF	83	96/4
7[d]	$FeCl_3{\cdot}6H_2O$ (5)	THF	88	98/2
8	$FeCl_3$ (10)	THF	61	95/5
9[e]	CuCl (10)	THF	0	—
10[f]	—	THF	0	—

[a] The reaction was conducted using 0.2 mmol of **1a**, 3 equiv. of MeMgCl, and the indicated amount of the corresponding catalyst in 3 mL of solvent. [b] NMR yield. [c] The reaction time was 1.0 h. [d] 1 mmol of **1a** was applied and the reaction time was 1 h. [e] Recovery: 80%. [f] Recovery: 77%.

The structure, as well as the relative configurations of the two chiral centers of **3a**, was unambiguously established by its single crystal X-ray diffraction analysis (Fig. 1),[20] which clearly indicated that the *in situ* generated metallic reagent undergoes an S_N2 substitution reaction at the β-position of 4-(trimethylsilylethynyl)oxetan-2-one affording 4-alkynoic acid **3** with the configuration of this carbon center inverted.

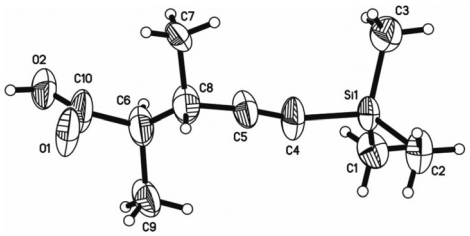
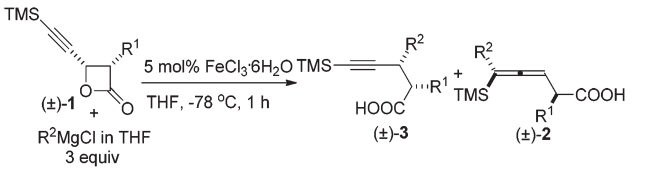


Fig. 1 ORTEP representation of the product **3a**.

Under this set of optimized reaction conditions, the scope and the limitations of this reaction were explored. The reaction of **1b** with MeMgCl afforded 2,3-dimethyl-5-(trimethylsilyl)-pent-4-ynoic acid **3b** with a yield of 84% and a selectivity of 97/3 (entry 1, Table 2). Aryl Grignard reagents could also be used in the reaction providing the corresponding products **3** in 73–89% yield with a regioselectivity of 93/7–97/3 (entries 2 and 5–8, Table 2). Different alkyl or allyl substituents could be introduced into the 2-position of the products in yields of 78%–87% with excellent ratios of 97/3–99/1 of **3**/**2** (entries 3 and 9–11, Table 2).

Table 2 Substrate scope of the $FeCl_3{\cdot}6H_2O$-catalyzed coupling reactions of racemic **1** with Grignard reagents[a]

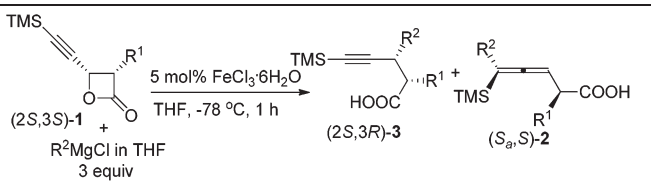


Entry	R^1	R^2	Yield of **3** (%)	**3**/**2**
1	H (**1b**)	Me	84 (**3b**)	97/3
2	H (**1b**)	Ph	80 (**3c**)	94/6
3	Me (**1a**)	Me	84 (**3a**)	98/2
4[b]	Me (**1a**)	Ph	88 (**3d**)	96/4
5	Me (**1a**)	*p*-MeC_6H_4	78 (**3e**)	95/5
6	Me (**1a**)	*m*-MeC_6H_4	79 (**3f**)	93/7
7	Me (**1a**)	*p*-$MeOC_6H_4$	73 (**3g**)	95/5
8	Me (**1a**)	2-Thienyl	89 (**3h**)	97/3
9[c]	CH_2CH_2Cl (**1c**)	Me	78 (**3i**)	97/3
10[d]	Allyl (**1d**)	Me	87 (**3j**)	99/1
11[c,e]	*n*-C_3H_7 (**1e**)	Me	77 (**3k**)	98/2

[a] The reaction was conducted using 1 mmol of **1**, 3 equiv. of Grignard reagents, and 5 mol% of $FeCl_3{\cdot}6H_2O$ in 5 mL of THF. [b] 0.4 mmol of **1a** was applied. [c] The reaction time was 1.5 h. [d] 0.5 mmol of **1d** was applied. [e] TES was used instead of TMS.

Based on these observations, we were anxious to see whether various optically active substrates **1** may be applied to this Fe-catalyzed coupling reaction. Interestingly, when optically active 4-(trimethylsilylethynyl)oxetan-2-ones were treated with different Grignard reagents, the corresponding optically active 4-alkynoic acids **3** were formed without any racemization (Table 3). Furthermore, we may easily conduct the reaction of 0.8976 g of (2*R*,3*R*)-**1g** and 0.8934 g of (2*S*,3*S*)-**1g** affording 0.9015 g of (2*R*,3*S*)-**3m** and 0.8967 g of (2*S*,3*R*)-**3m** in excellent yield and selectivity (entries 7 and 8, Table 3).

Table 3 Substrate scope of the diastereoselective transformation of β-lactones **1** to 4-alkynoic acids **3**[a]



Entry	**1** R^1 (ee%)	R^2	Yield of **3** (%) (ee%)	**3**/**2**
1	(*S*,*S*)-**1a** Me (98)	Me	81 (99) (*S*,*R*)-**3a**	98/2
2	(*S*,*S*)-**1a** Me (98)	Ph	77 (98) (*S*,*R*)-**3d**	95/5
3[b]	(*S*,*S*)-**1a** Me (98)	PMP	72 (97) (*S*,*R*)-**3g**	95/5
4	(*R*,*R*)-**1d** allyl(99)	Me	81 (98) (*R*,*S*)-**3j**	99/1
5	(*S*,*S*)-**1f** ^{n}Pr (97)	Me	74 (97) (*S*,*R*)-**3l**	98/2
6	(*R*,*R*)-**1f** ^{n}Pr (>99)	Me	73 (>99) (*R*,*S*)-**3l**	97/3
7[c]	(*S*,*S*)-**1g** Me(>99)	Me	94 (99) (*S*,*R*)-**3m**	98/2
8[c]	(*R*,*R*)-**1g** Me(>99)	Me	94 (98) (*R*,*S*)-**3m**	99/1

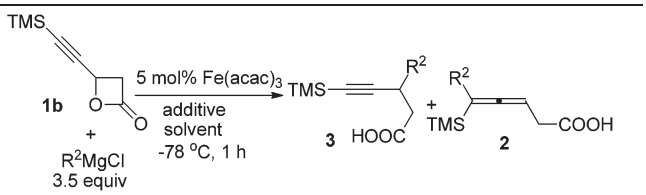
[a] The reaction was conducted using 1 mmol of **1**, 3 equiv. of Grignard reagents, and 5 mol% of $FeCl_3{\cdot}6H_2O$ in 5 mL of THF. [b] The reaction time was 2 h. [c] TBS was used instead of TMS and 4 mmol of the **1g** were applied.

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However, the reaction of $FeCl_3·6H_2O$-catalyzed coupling of **1b** bearing just one chiral center with *n*-C_4H_9MgCl afforded a complex mixture (entry 1, Table 4) while a ratio of 75/25 of **3**/**2** was observed when $Fe(acac)_3$ was used instead of $FeCl_3·6H_2O$ (entry 2, Table 4). Et_2O appeared better than THF (entry 3, Table 4) and the addition of 1 equiv. of NaI, added as ligand, also helped (entry 4, Table 4). By comparing the results with these shown in Table 3, we reasoned that water from $FeCl_3·6H_2O$ may have a unique role. In fact, the addition of 0.3 equiv. of H_2O did further improve the regioselectivity of this coupling reaction using $Fe(acac)_3$ as the catalyst (entries 4 and 5, Table 4). The reactions of **1b** with *n*-C_4H_9MgCl and *n*-$C_5H_{11}MgCl$ afforded the corresponding products **3n** and **3o** in 71% and 69% yields with an excellent ratio of 94/6 of **3**/**2** (entries 5 and 6, Table 4).

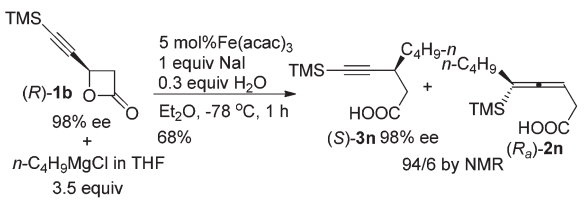
Table 4 $Fe(acac)_3$-catalyzed coupling reactions of **1b** with alkyl Grignard reagents[a]



Entry	R^2MgCl	Solvent	Additive	Yield of **3**[b] (%)	**3**/**2**
1[c]	*n*-C_4H_9	THF	—	Complex mixture	
2	*n*-C_4H_9	THF	—	60 (**3n**)	75/25
3	*n*-C_4H_9	Et_2O	—	70 (**3n**)	89/11
4	*n*-C_4H_9	Et_2O	NaI	64 (**3n**)	91/9
5[d]	*n*-C_4H_9	Et_2O	NaI–H_2O	71 (**3n**)	94/6
6[d]	*n*-C_5H_{11}	Et_2O	NaI–H_2O	69 (**3o**)	94/6

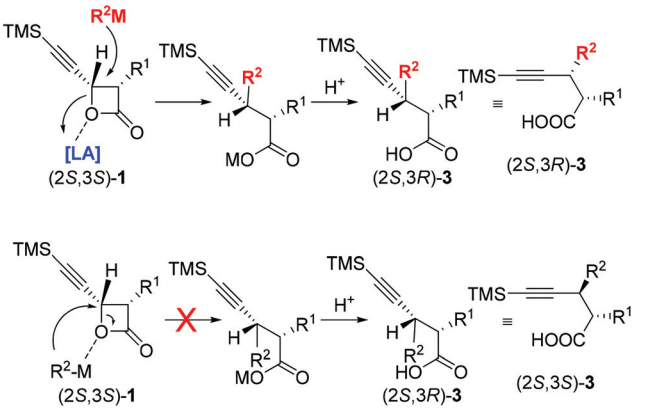
[a] The reaction was conducted using 0.2 mmol of **1b**, 3.5 equiv. of the Grignard reagent, 5 mol% of $Fe(acac)_3$, 1 equiv. of NaI (if required), 0.3 equiv. H_2O (if required), 2 mL of solvent. [b] Isolated yield of **3**. [c] $FeCl_3·6H_2O$ was applied. [d] 1 mmol of **1b** was applied.

Furthermore, we conducted the S_N2 substitution reaction of (*R*)-**1b** with a single chiral center with *n*-C_4H_9MgCl affording (*S*)-**3n** without any racemization in 68% yield with a ratio of 94/6 of (*S*)-**3n**/(R_a)-**2n** (eqn (3)).



(3)

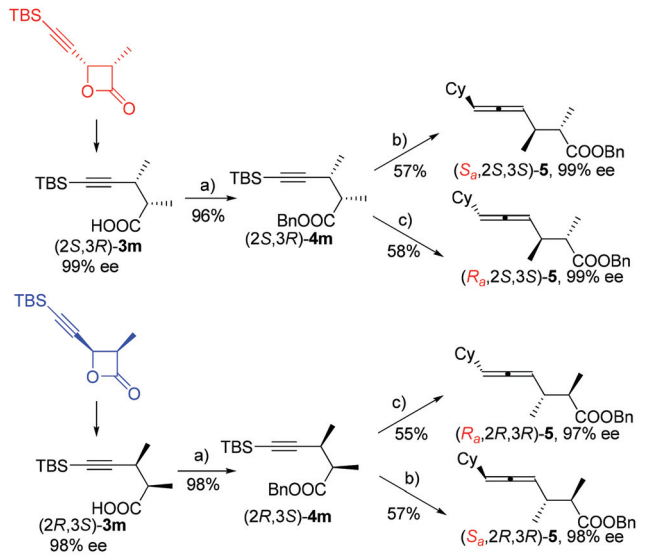
These "inverted" observations led to the conclusion that the mechanism is different from the reported one under Fe-catalysis with the coordination-directed direct delivery of the R group[16] from the *in situ* generated highly reduced iron–magnesium cluster reagent.[17] We believed that the attack of the R^2 from *in situ* generated highly reduced iron–magnesium cluster reagent R^2–M at the 4-position of 4-(trimethylsilyl)ethynyloxetan-2-one (2*S*,3*S*)-**1** may be mediated by the Lewis acidic species in the solution coordinating with the oxygen atom to afford (2*S*,3*R*)-**3**. The steric effect of the trimethylsilylacetylenyl group is beneficial to this transformation. The configuration of the chiral center in the products (2*S*,3*R*)-**3** followed the Walden inversion which shows that the coupling reaction proceeds *via* an S_N2 mechanism with complete inversion of the carbon center (Scheme 2).[21]



Scheme 2 Rationale for the inverted absolute configuration.

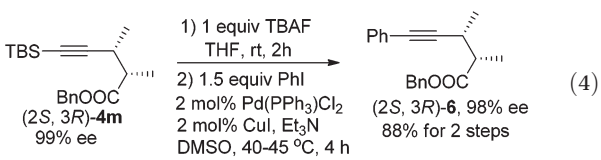
It is well established that the extra C–C triple bond in the products is synthetically very useful in organic synthesis, undergoing many reactions[22] such as highly stereoselective ionic additions, semi-hydrogenation, complete hydrogenation, and manipulation based on the terminal acidic C–H bond. In addition to these known protocols, here, (2*S*,3*R*)-5-(*t*-butyldimethylsilyl)-2,3-dimethylpent-4-ynoic acid (2*S*,3*R*)-**3m** was converted to the benzyl ester easily, which was followed by desilylation by treating with TBAF. This terminal alkynoic acid ester may undergo an enantioselective allenylation of terminal alkyne reaction (EATA) with (*R*)- or (*S*)-diphenylprolinol and CyCHO[23] providing (S_a,2*S*,3*S*)-benzyl-6-cyclohexyl-2,3-dimethylhexa-4,5-dienoate (S_a,2*S*,3*S*)-**5** or (R_a,2*S*,3*S*)-benzyl-6-cyclohexyl-2,3-dimethylhexa-4,5-dienoate (R_a,2*S*,3*S*)-**5** in excellent de and ee (Scheme 3). The absolute configuration of the allene unit has been assigned based on our previous report.[23] Likewise, its enantiomers (S_a,2*R*,3*R*)-**5** and (R_a,2*R*,3*R*)-**5** could be prepared similarly by using (2*R*,3*S*)-5-(*t*-butyldimethylsilyl)-2,3-dimethylpent-4-ynoic acid (2*R*,3*S*)-**3m** as the starting material.

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Scheme 3 Synthesis of four diastereoisomers of **5** with two asymmetric carbon centers and one axially chiral allene. (a) 1.5 equiv. BnBr, 3 equiv. $NaHCO_3$, DMF, 30 °C, 16 h. (b) i. 1 equiv. TBAF, THF, rt, 2 h; ii: 1.8 equiv. CyCHO, 0.8 equiv. $ZnBr_2$, 1.25 equiv. (*R*)-diphenylprolinol, toluene, 120 °C, 24 h. (c) i. 1 equiv. TBAF, THF, rt, 2 h; ii. 1.8 equiv. CyCHO, 0.8 equiv. $ZnBr_2$, 1.25 equiv. (*S*)-diphenylprolinol, toluene, 120 °C, 24 h.

In addition, (2*S*,3*R*)-**4m** may undergo a desilylation reaction followed by a Sonogashira coupling[24] with PhI to afford (2*S*,3*R*)-benzyl 2,3-dimethyl-5-phenylpent-4-ynoate (2*S*,3*R*)-**6**, which further extends the potential of this reaction by introducing an extra substituent at this location (eqn (4)).



(4)

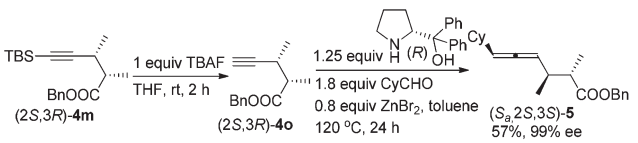
In summary, we have achieved an Fe-catalyzed highly regio- and stereoselective coupling reaction of 4-ethynyl-oxetan-2-ones with different Grignard reagents affording 4-alkynoic acids in good yields with the absolute configuration of the β-carbon atom completely inverted. The optically active acids may be easily esterified, which may be followed by desilylation to unveil the attractive terminal C–C triple bond for arylation or enantioselective allenylation (EATA) to afford optically active 4,5-allenoates with defined advanced steric nature. Although the mechanism, especially the real nature of the *in situ* generated metallic species and the role of NaI as well as water, is far from clear, due to the easy availability of the Fe-catalyst and optically active starting materials,[25] high regio- and stereoselectivity, and potential of the products, this reaction will be useful in organic synthesis involving multiple chiralities. Further studies including the effect of water in this area are ongoing in our laboratory.

Experimental

1. Fe-catalyzed S_N2 coupling reaction of Grignard reagent with 1. Synthesis of 2,3-dimethyl-5-(trimethylsilyl)pent-4-ynoic acid 3a

Typical procedure. To a mixture of $FeCl_3{\cdot}6H_2O$ (13.7 mg, 0.05 mmol), **1a** (183.0 mg, 1 mmol), and THF (5 mL) was added dropwise a solution of MeMgCl (1 mL, 3 M in THF, 3 mmol) at − 78 °C within 3 min under N_2 atmosphere. After being stirred at −78 °C for 1 h, the reaction mixture was quenched with EtOH (0.5 mL), and then acidified with 5% HCl (aq.) to pH = 1. The resulting mixture was extracted with ether (15 mL × 3), washed with brine, filtered, and evaporated. The ratio of **3a**/**2a** (= 98/2) was determined by ^{1}H NMR analysis of the crude reaction mixture before separation. The residue was purified by flash chromatography on silica gel (eluent: petroleum ether–ethyl acetate = 10 : 1–2 : 1 to afford **3a** (167.3 mg, 84%): Solid: m.p. 67.3–68.4 °C (hexane–ethyl acetate); ^{1}H NMR (300 MHz, $CDCl_3$) δ 10.54 (brs, 1 H, COOH), 2.83 (pentet, J = 7.2 Hz, 1 H, CH), 2.42 (pentet, J = 7.2 Hz, 1 H, CH), 1.32 (d, J = 6.9 Hz, 3 H, CH_3), 1.22 (d, J = 6.9 Hz, 3 H, CH_3), 0.14 (s, 9 H, 3 × CH_3Si); ^{13}C NMR ($CDCl_3$, 75 MHz) δ 181.6, 108.0, 86.6, 45.1, 30.0, 19.4, 14.8, 0.1; IR (neat, cm^{-1}) 2977, 2938, 2899, 2169, 1712, 1460, 1427, 1373, 1268, 1245, 1212, 1088; MS (EI) m/z (%) 198 (M^+, 0.23), 183 ($(M - CH_3)^+$, 20.67), 75 (100); Elemental analysis calcd for $C_{10}H_{18}O_2Si$: C, 60.56, H, 9.15, found: C, 60.40, H, 8.95%.

2. Desilylation and enantioselective allenylation of 4m. Synthesis of (S_a,2*S*,3*S*)-benzyl 6-cyclohexyl-2,3-dimethylhexa-4,5-dienoate (S_a,2*S*,3*S*)-5



Typical procedure. To a solution of (2*S*,3*R*)-**4m** (164.5 mg, 0.50 mmol) in THF (3 mL) was added TBAF (0.5 mL, 1 M in THF, 0.5 mmol). After stirring for 2 h at rt, the resulting solution was quenched with water (5 mL), extracted with ether (15 mL × 3), washed with brine, dried over Na_2SO_4, filtered, and evaporated. The residue was purified by flash chromatography on silica gel (eluent: petroleum ether–ethyl ether = 60 : 1) to afford (2*S*,3*R*)-benzyl 2,3-dimethylpent-4-ynoate (2*S*,3*R*)-**4o**, which was used directly in the next step.

To a reaction tube was added $ZnBr_2$ (90.3 g, 0.40 mmol). This reaction tube was then dried under vacuum with a heating gun. (*R*)-Diphenylprolinol (156.4 mg, 0.62 mmol), CyCHO (101.2 mg, 0.90 mmol)/toluene (1 mL), and (2*S*,3*R*)-benzyl 2,3-dimethylpent-4-ynoate (2*S*,3*R*)-**4o**/toluene (2 mL) were then added sequentially under a N_2 atmosphere. The reaction tube was then placed in a pre-heated oil bath at 120 °C. After the reaction was complete as monitored by TLC, the reaction mixture was cooled to rt and the crude reaction

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mixture was filtered through a short pad of silica gel (2 cm) eluted with ether. After evaporation, the residue was purified by flash chromatography on silica gel (petroleum ether–ethyl ether = 80 : 1) to afford (S_a,2*S*,3*S*)-**5** (88.5 mg, 57%, 99% ee: HPLC conditions: OJ-H column, rate = 0.7 mL min^{-1}, eluent: hexane–i-PrOH = 400 : 1, λ = 214 nm, t_R 14.8 min (minor), 15.8 min (major)): Liquid; $[\alpha]_D^{20}$ = +11.6 (*c* = 1.82, $CHCl_3$); ^1H NMR (300 MHz, $CDCl_3$) δ 7.40–7.30 (m, 5 H, ArH), 5.19–5.00 (m, 4 H, CH_2 + 2 × CH═), 2.58–2.35 (m, 2 H, 2 × CH), 2.01–1.85 (m, 1 H, CH), 1.80–1.56 (m, 5 H, 5 protons of Cy), 1.35–0.95 (m, 11 H, 5 protons of Cy + 2 × CH_3); ^{13}C NMR ($CDCl_3$, 75 MHz) δ 202.4, 175.6, 136.0, 128.5, 128.1, 98.2, 94.2, 66.0, 45.5, 37.1, 36.6, 33.04, 33.00, 26.1, 26.0, 18.4, 14.1; IR (neat, cm^{-1}) 2966, 2925, 2851, 1959, 1735, 1498, 1449, 1379, 1345, 1254, 1219, 1158; MS (EI) *m*/*z* (%) 312 (M^+, 1.10), 165 (100), 91 (100), 69 (100); HRMS calcd for $C_{21}H_{28}O_2$ (M^+): 312.2089, found: 312.2092.

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